



OPEN Exploration of brain function changes in the visual cortex of astigmatic subjects based on fNIRS

Yuling Tang^{1,3}, Chang Wu^{1,3}, Mengtian Bai¹, Qingqing Tan¹, Jing Tian¹, Jia Lin¹, Xiangyu Chen¹, Mengxiong Luo¹, Zongwei Chen¹, Yi Shao²✉ & Xuan Liao¹✉

The primary objective of this research is to explore changes in brain activity among astigmatism patients, employing functional near-infrared spectroscopy (fNIRS). A prospective cross-sectional study comprising 57 participants (99 eyes) was conducted and segmented into three distinct sections. In PART I, 24 participants (48 eyes) with astigmatism (Group A) and 18 participants (36 eyes) without astigmatism (Group B) were assessed using fNIRS to investigate the variations across different brain regions and channels. PART II involved 15 participants with monocular astigmatism (15 eyes), who received complete spherical optical correction, followed by fNIRS assessments after visual stimulation using E-Prime software. The study examined the activation levels of the occipital visual cortex and the functional connectivity strength of the brain, comparing these metrics pre- and post-cylindrical correction. PART III: Following the correction of the initial refractive error in Group A, +2.00 diopter cylinder (DC) lenses were employed to induce astigmatism at various axes. These axes were categorized into three groups: oblique astigmatism group, with-the-rule astigmatism (WTR), and against-the-rule astigmatism (ATR). fNIRS was conducted during task performance to assess the strength and activation level of brain functional connectivity across different astigmatism axes of the same diopter. PART I: The functional connectivity strength of HbO₂-based occipital cortex was stronger in group B than in group A ($P < 0.001$). PART II: Significant differences in the β values of the left and right visual cortex were detected in cases of monocular astigmatism before and after astigmatism correction. ($P = 0.002$ and $P = 0.004$). The functional connectivity strength in the post-cylindrical correction condition exceeded that in the pre-cylindrical correction condition; nevertheless, this variation did not attain statistical significance ($P > 0.05$). PART III: The functional connectivity strength and activation level of the HbO₂-based occipital visual cortex exhibited significant variations across different astigmatism axes ($P < 0.001$ and $P = 0.045$). Specifically, the functional connectivity strength in the oblique astigmatism group was greater than that observed in both the WTR and ATR groups. Conversely, the degree of brain activation in the oblique astigmatism group was lower compared to the ATR group ($P = 0.042$). These observations imply that astigmatism may decrease both the activation of the occipital visual cortex and the strength of brain functional connectivity. The functional activities of the occipital cortex can be enhanced to a certain degree following optical correction of astigmatism. The strength of functional connectivity and the degree of activation in the occipital visual cortex vary with different axes of astigmatism.

Trial Registration This study was registered in the China Clinical Trials Registry (registration number: ChiCTR2300070613, date of registration: 2023-04-18), and the study followed the ethical principles required by the Declaration of Helsinki, and the subjects were informed and signed the informed consent form, and it has been reviewed and approved by the Medical Ethics Committee of the Affiliated Hospital of North Sichuan Medical College (Approval No. 2022ER541-1).

Keywords Astigmatism, fNIRS, Visual cortex, Activation, Strength, Brain functional connectivity

¹Department of Ophthalmology of Affiliated Hospital and Ophthalmology & Optometry of Medical School, North Sichuan Medical College, Nanchong 637000, Sichuan Province, China. ²Department of Ophthalmology of Shanghai General Hospital, National Clinical Research Center for Eye Disease, Shanghai Jiao Tong University School of Medicine, Shanghai 200080, China. ³Yuling Tang and Chang Wu have contributed equally to this work. email: aleexand@163.com; freebee99@163.com ✉ email: freebee99@163.com; aleexand@163.com

Astigmatic errors differ from myopic and hyperopic ones, because differential ocular surface curvature causes inability to focus image points as single points on the retina; this results in blurred vision at any distance¹. Approximately 20–40% of the population have at least 1 diopter (D) of refractive astigmatism². However, when lower magnitudes of astigmatism are included, the prevalence can exceed 60%³. Research indicates that the prevalence of astigmatism rises with age. The prevalence of astigmatism > 1.00 DC is approximately 8.30–11.60% in individuals aged 5–30 years, 15.70–21.00% in those aged 30–60 years, and 31.10–81.80% in those aged 60–90 years⁴. Patients with astigmatism can adapt to projected minimal diffuse spots onto the retina, thereby enhancing unaided vision. However, sustained accommodation may lead to increased fatigue, and uncorrected astigmatism can adversely affect visual quality⁵. Despite the correction of astigmatism, visual acuity (VA) remains significantly diminished in certain meridian directions. Even when employing laser interferometry, which bypasses the optical system of the eye to project sinusoidal fringes directly onto the retina, the measured retinal VA continues to exhibit variability across different meridians. This phenomenon suggests that the underlying mechanism may involve the visual cortex⁶. Clinical evidence dating back to 1969 indicates that astigmatism can be present at birth in infants and young children, and when astigmatism exceeds 0.75 DC, it can cause symptoms such as blurred vision and diplopia⁷. In terms of animal model studies, myopic astigmatism alters the electrophysiological responses of the retina in chickens, particularly delaying the inner retinal responses in the central region. The influence of astigmatism on retinal responses exhibits region-specific characteristics and correlates with the axis of astigmatism⁸, and astigmatism is also a major risk factor for amblyopia⁹. One year following cylindrical intervention for astigmatism in school-aged children, their visual function (mainly spherical aberration) remains inferior to that of children without astigmatism¹⁰. Furthermore, a reduction in the amplitude of visual electrophysiological responses is evident in children with astigmatism¹¹. This suggests that visual performance is influenced by both the ocular optical system and the response of the central visual nervous system¹². Consequently, optical correction of astigmatism alone may not suffice to achieve optimal visual outcomes, potentially implicating the developmental interplay of the brain's central visual pathways.

Prior research has demonstrated that refractive errors significantly impact the functional activity of the visual cortex. Visual electrophysiological investigations have indicated that myopic defocus can result in a reduction of the P100 amplitude and an extension of its latency^{13–15}. Functional magnetic resonance imaging (fMRI) and fNIRS provide direct evidence of alterations in hemodynamic responses, as well as changes in brain structure and function, in individuals with high myopia^{16,17}. Freeman et al.¹⁸ discovered that individuals with astigmatism exhibited differences in resolution between the vertical and horizontal meridians, even after refractive correction.; Numerous subsequent studies have documented that individual with high astigmatism exhibit direction-specific visual deficits even following optical correction after the critical period of visual development^{10,19}. During this critical period, abnormal visual input resulting from uncorrected astigmatism can adversely impact the development of spatial vision, potentially leading to the onset of meridian amblyopia²⁰. Furthermore, neurological deficits in the V1 region of the visual cortex have been observed in both animal experiments and human psychophysical studies investigating meridian amblyopia induced by astigmatism²¹.

Research involving animal models has shown that early abnormalities in visual input can significantly impact the neural tissue within the mammalian visual system, with the effects of such exposure being permanent and irreversible even after the restoration of normal visual input^{6,12}. The majority of neurokinines influencing binocular visual development are located in the V1 region, where neural connections commence formation shortly after birth. The maturation of visual processing and neural connections is contingent upon early visual experiences. Prolonged exposure to distorted visual input during critical periods of visual development may adversely affect normal visual processing and the development of neural connections²². Specifically, abnormal visual experiences resulting from astigmatism can disrupt the development of functional connectivity in the visual cortex, which subsequently manifests in perceptual behavior during adulthood. In comparison to individuals without astigmatism, adults with astigmatism exhibit a significant distortion in shape perception, which is only partially rectified through optical correction aimed at improving distorted visual input²³. These discoveries suggest a connection between astigmatism and the nervous system, implicating the visual cortex as a key component in the underlying mechanism. However, current research on the mechanisms underlying astigmatism predominantly focuses on the optical system^{24–26}, with insufficient systematic investigation into its effects on central visual function.

fNIRS is a non-invasive brain imaging technique that utilizes the unique absorption properties of near-infrared light (600–900 nm wavelength) by hemoglobin in cerebral tissue, enabling real-time detection of blood oxygenation dynamics in the cerebral cortex²⁷. Due to its non-invasive nature and capability for continuous monitoring, this approach provides an innovative and straightforward method for analyzing, diagnosing, and treating diseases in clinical practice. Studies have found that fNIRS demonstrates good sensitivity and repeatability in detecting cognitive load and states, suggesting that fNIRS is well-suited for exploring the full complement of neuroimaging research questions and can serve as a viable alternative to fMRI^{28,29}. At present, it is predominantly applied in brain diseases, psychiatric problems, cognitive fields and sports rehabilitation, but there are few related studies in the field of ophthalmic diseases, with only a few reports on strabismus amblyopia^{30,31}, diabetic retinopathy³², pathological myopia³³, etc. fNIRS serves as a valuable complement to existing modalities such as fMRI, positron emission tomography (PET), and electroencephalography (EEG). The near-infrared light emitted by fNIRS can penetrate the skull and reach the cerebral cortex to a depth of approximately 2–3 cm. Within the cortex, the light undergoes scattering and propagation before exiting the skull, where it is detected by a collection device. The intensity of the collected near-infrared light varies with the concentration of blood oxygen in the cerebral cortex. An increase in neural activity within a specific brain region leads to a corresponding rise in cerebral blood flow, which is reflected by alterations in hemoglobin levels. This relationship between cortical activation and blood flow is referred to as neurovascular coupling³⁴. In comparison to fMRI, which relies on magnetic fields, fNIRS detects alterations in hemoglobin concentration through optical

principles. It offers superior temporal resolution and is more accessible for a broader demographic due to its user-friendly operation and portability. When contrasted with PET, fNIRS is non-invasive, posing fewer risks to the human body, and allows for repeated measurements. Additionally, compared to EEG, fNIRS provides enhanced spatial resolution, facilitating a more precise correlation between anatomical structures and brain function.

Given the limited research on alterations in brain function within the visual cortex of individuals with astigmatism, this investigation employed fNIRS to examine differences in the strength of brain connectivity between individuals with and without astigmatism. Additionally, the research aimed to identify changes in brain function before and after optical correction in participants with moderate and low astigmatism, and to assess potential differences in activation levels between the left and right visual cortex in individuals with monocular astigmatism. An additional aim of this investigation is to examine the variations in alterations of brain function associated with different axes of astigmatism, with the aim of elucidating the central mechanisms underlying astigmatism and generating insights for enhancing the visual quality of individuals affected by this condition.

Patients and methods

Patients

A cohort of 60 healthy young individuals who underwent routine ophthalmic examinations at the outpatient clinic of the Affiliated Hospital of North Sichuan Medical College between May 2023 and October 2023 was initially selected for this study. However, one participant was unable to complete the entire experimental procedure, and two were excluded due to poor signal acquisition quality. Thus, the final sample consisted of 57 participants. Among these, 39 presented with astigmatism, including 15 individuals with monocular astigmatism (15 eyes) and 24 individuals with binocular astigmatism (48 eyes). The remaining 18 participants had no astigmatism (36 eyes) and served as the control group. Astigmatism was determined using an autorefractor and subjective refraction made with a phoropter. Meanwhile, astigmatism correction was performed using trial frame spectacles. For Part III, based on the correction of each participant's original refractive error, +2.00 D cylindrical lenses in a trial frame were used to induce astigmatism of the same diopter but with different axes.

Inclusion criteria: (1) aged 18–30 years, with best corrected VA of 0.0 (Log MAR) in both eyes; (2) axial length < 26 mm; (3) spherical power > −6.00 DS; (4) cylindrical power between −0.50 DC and −2.00 DC (for the astigmatism groups); (5) cylindrical power less than 0.50 DC (for the non-astigmatic group). Exclusion criteria: (1) Eye diseases such as pathological myopia, glaucoma, and corneal disease that affect vision and visual quality; (2) history of any ocular surgery or trauma; (3) Those who cannot cooperate with the inspection process.

The requisite sample size for this investigation was ascertained utilizing PASS 15.0 software, with a significance threshold defined at $\alpha = 0.05$ and a test power $(1 - \beta)$ of 0.9. Based on the estimated sample rate for comparing the means of a t-test between two independent samples, the total sample consisted of 32 cases. According to the sample rate estimation of the paired sample t-test mean comparison, the total sample size was 11 cases. Based on the sample rate estimate of one-way analysis of variance (ANOVA) mean comparison, the total sample comprised 30 cases. The overall sample size for this study comprised 57 cases, thereby providing a statistically significant basis for analysis.

fNIRS

Previous studies have demonstrated that fNIRS has excellent reproducibility, reliability and validity in the assessment of brain function^{35–37}. The variation in hemoglobin concentration within the visual cortex of the occipital lobe was continuously monitored using a near-infrared spectroscopy based functional brain imaging apparatus. (Danyang Huichuang Medical Equipment Co., Ltd., Jiangsu, China). The system is comprised of a near-infrared light source, specifically a light-emitting diode (LED), and an avalanche photodiode (APD) functioning as detectors. The light source probes operate at wavelengths of 760 nm and 850 nm, individually, with a sampling rate of 11 Hz. In the experimental setup, four light source probes and four detectors were utilized, resulting in eight effective channels. Channels CH01 to CH04 were designated for the right occipital lobe, while channels CH05 to CH08 were allocated to the left occipital lobe. The distance between the transmitter and the detector varies from 2.7 to 3.3 cm, with both components secured using a black head cap. HbO₂ is the most sensitive marker for changes in local cerebral blood flow and is therefore the most effective marker for reflecting changes in brain function. Consequently, this study primarily focuses on the analysis of HbO₂ variations.

Data collection

During the resting state data collection, the specialized examination room is maintained in a quiet environment. The participant is seated comfortably in a chair and fitted with a black head cap, which is arranged following the International 10–20 system. The central electrode is aligned with the occipital protuberance, and the remaining electrodes are securely attached to the scalp. The duration of the test is 8 min, during which the participant is instructed to refrain from speaking and to minimize head and body movements. During the task state data collection, the experimental conditions mirrored those of the head-mounted apparatus. Each measurement was performed once. The time interval between correction or induced astigmatism and experimental data collection was similar across all participants. Visual stimulation tasks were administered and data were recorded using E-Prime 3.0 software. Resting-state data acquisition was employed in Part I, whereas task-based data acquisition was adopted in Parts II and III. The flowchart of the methodology is shown in Fig. 1.

Visual stimuli: In this study, the standard pattern of 1 CPD was inverted by using Photoshop mapping software to draw a standard pattern inversion checkerboard of visual evoked potentials recommended by the International Society of Clinical Electrophysiology³⁸. A red cross was positioned at the center of the checkerboard as a fixation point. The inversion stimulus was administered at a temporal frequency of 2.0 Hz on a 15.6-inch monitor with a resolution of 1920 × 1080 pixels. As illustrated in Fig. 2, the overall duration of stimulation was 140 s. During

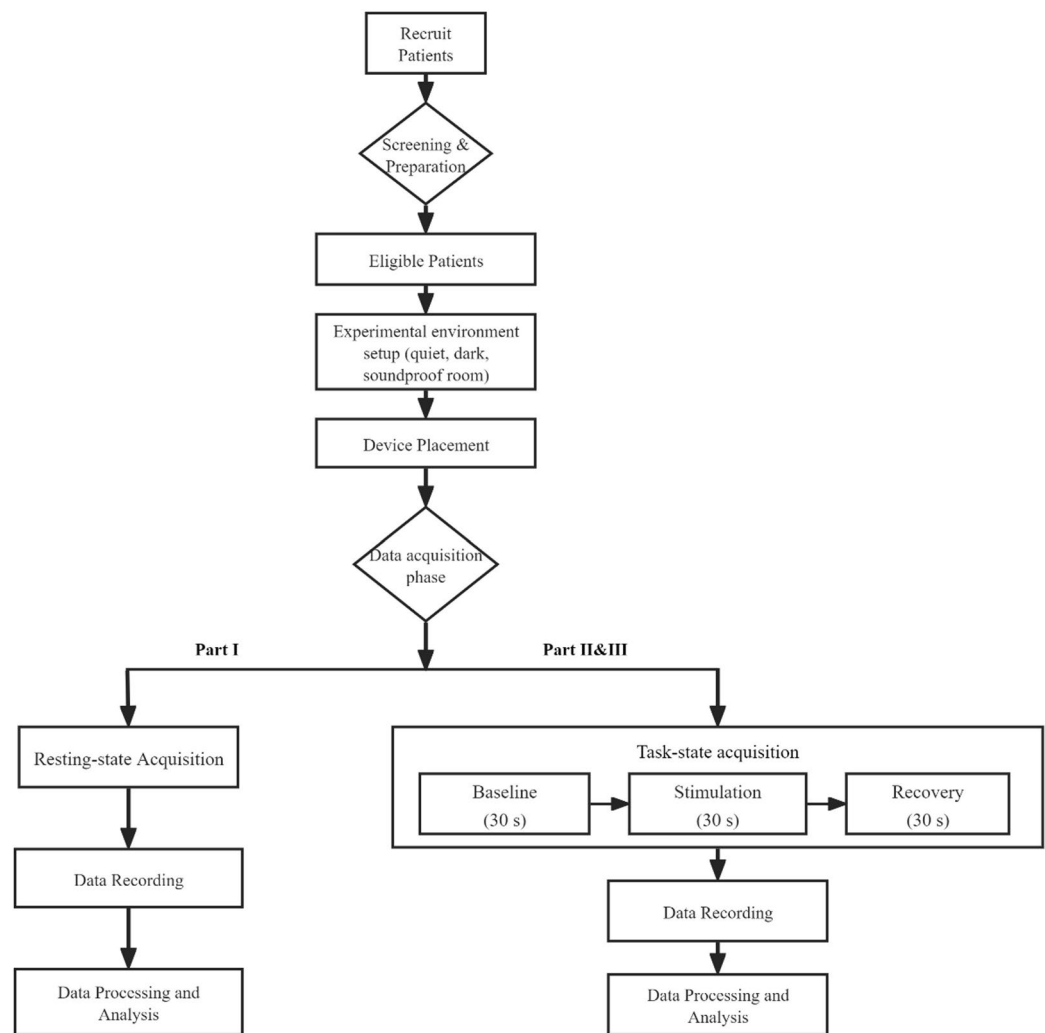


Fig. 1. Method flowchart.

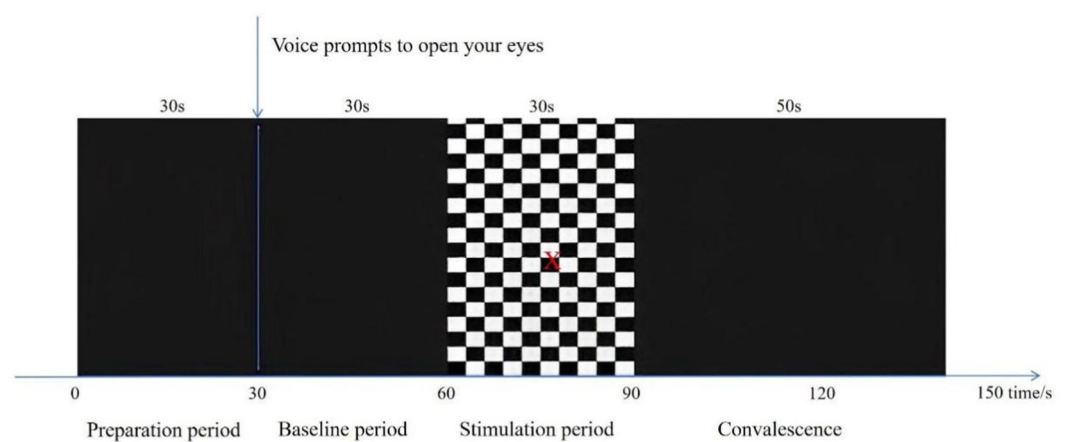


Fig. 2. Programming of visual stimulation test. The red cross in the center indicates the fixation point.

the 30-s preparation phase, participants were instructed to close their eyes and subsequently open them upon hearing auditory prompts. The viewing durations were allocated as follows: 30 s for the baseline period, 30 s for the visual stimulation period, and 50 s for the recovery period.

Data processing and statistical analysis

The initial spectral data collected were preprocessed using NirSpark software. Temporal regions not pertinent to the experiment were excluded, and motion artifacts and filtering were removed. Subsequently, changes in blood oxygen concentration were calculated in accordance with the modified Beer-Lambert law. The functional connectivity strength and β value for each channel were analyzed, with the β value employed to assess the activation degree of the visual cortex. Statistical analyses were carried out with SPSS software, version 26.0. The Shapiro–Wilk test utilized to evaluate the conformity of the data to a normal distribution. Data conforming to a normal distribution were expressed as the mean $\pm$ standard deviation ($\bar{x} \pm s$), while data deviating from a normal distribution were represented by the median and interquartile range [M (Q1, Q3)]. Disparities in age, intraocular pressure, and gender across the two groups were evaluated employing independent samples t-tests and chi-square tests. Specifically, the independent samples t-test was utilized to evaluate differences in the strength of brain functional connectivity between groups A and B. In the second section of the analysis, a paired sample t-test was employed to evaluate the differences in brain functional connectivity strength and β values between groups C and D, as well as to assess the disparities in β values between the left and right occipital visual cortices. Subsequently, in the third section, a repeated measures ANOVA was conducted to examine the variations in brain functional connectivity strength and β values across different axial groups within the same diopter. The Bonferroni method was utilized for pairwise comparisons. A p-value below 0.05 was deemed indicative of statistical significance.

Results

Differences in brain function connectivity characteristics and connection strength in patients with or without astigmatism in the resting state

The initial phase of the study comprised 42 participants (84 eyes), including 24 individuals (48 eyes) with binocular astigmatism and 18 individuals (36 eyes) without astigmatism. Baseline data for these subjects are presented in Table 1. Participants with astigmatism were categorized as Group A, while those without astigmatism were designated as Group B. Figure 3 illustrates the average brain functional connectivity strength of the occipital lobe for groups A and B, as measured by HbO₂ levels. The data indicate that group B exhibits a greater and more dispersed connectivity strength compared to group A, which demonstrates a smaller and more concentrated connectivity strength. Notably, the functional connectivity strength in group A is significantly weaker than that in group B, with the difference being statistically significant ($P < 0.001$).

As illustrated in Figs. 4 and 5b, statistically significant differences between the two groups were observed between channels CH01 and CH05 ($P = 0.040$) and between channels CH01 and CH06 ($P = 0.013$). In contrast, the remaining channels did not exhibit statistical significance ($P > 0.05$). Figure 5b demonstrates a statistically significant difference in functional connectivity strength between the two groups within the right occipital lobe brain network ($P = 0.004$). Nevertheless, no significant disparity was observed between the groups within the left occipital lobe brain network ($P = 0.818$). A significant statistical divergence was observed in the strength of brain functional connectivity between the two groups. ($P = 0.001$).

Comparison of brain function changes and activation degree of left and right visual cortex before and after correction of astigmatism in the task state

The second part of the research included a total of 15 subjects (15 eyes) with astigmatism in the right eye, consisting of 7 males (7 eyes) and 8 females (8 eyes), with an average age of 22.08 ± 2.05 years. These subjects had spherical lens prescriptions of -3.68 ± 0.58 DS and cylindrical lens prescriptions of -1.00 ± 0.24 DC. Participants underwent visual stimulation of the right eye both prior to and following astigmatism correction, with the left eye occluded. The use of spherical lenses for myopia correction and cylindrical lenses for astigmatism correction categorized participants into pre- and post-cylindrical correction conditions.

In comparison to the pre-cylindrical correction condition, the post-cylindrical correction condition exhibited enhanced functional connectivity within the left occipital lobe homology brain network and the bilateral occipital lobe heterologous brain network, alongside reduced connectivity in the right occipital lobe homology

	Binocular astigmatism subject group	Group of subjects without astigmatism	P-value
Gender/n			0.7
Male	11	10	
Female	13	8	
Eyes/cases	48	36	
Age/years	21.08 ± 2.64	23.00 ± 3.19	0.12
Intraocular pressure/ mmHg	15.92 ± 2.23	14.83 ± 2.37	0.26
Spherical degree/ DS	-3.59 ± 1.16	-3.45 ± 1.36	0.64
Cylindrical degree/ DC	-1.29 ± 0.43	-0.17 ± 0.24	<0.001

Table 1. Comparison of baseline data.

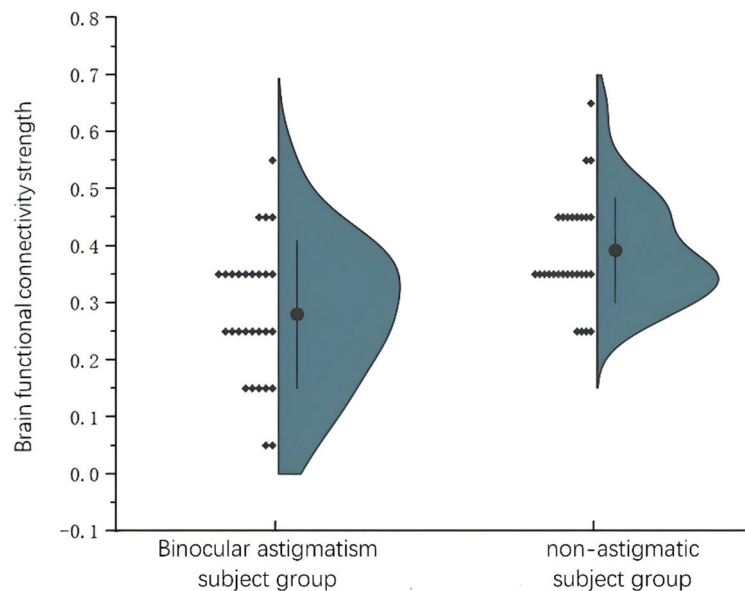


Fig. 3. Based on the characteristic distribution of average brain functional connectivity intensity within the occipital visual cortex of the HbO_2 brain, participants were categorized into two groups: Group A, comprising individuals with binocular astigmatism, and Group B, consisting of individuals without astigmatism. The small dots on the left illustrate the numerical distribution of brain functional connectivity intensity across each connecting channel, while the middle dot denotes the mean value, and the black vertical line represents the mean ± 1 standard deviation (SD).

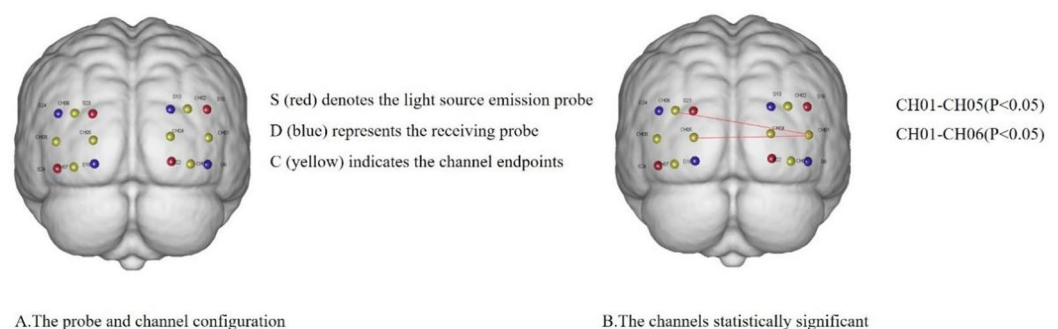


Fig. 4. The probe and channel configuration (A) and the channels statistically significant (B): The channels demonstrating statistically significant differences between individuals with binocular astigmatism and those without astigmatism were identified as CH01-CH05 and CH01-CH06 ($P < 0.05$).

brain network. However, the aggregate disparity in the strength of brain functional connectivity across the two cohorts did not attain statistical significance ($P = 0.929$). During the visual stimulation task involving the right eye, a comparative analysis of the β values for each channel of the occipital lobe of the HbO_2 brain in pre- and post-cylindrical correction conditions is presented in Fig. 6. Notably, significant differences were observed in the β values for channels CH02 ($P = 0.036$), CH06 ($P = 0.014$), and CH08 ($P = 0.024$) in post-cylindrical correction condition, whereas no significant differences were detected in the β values of the other channels. Table 2 illustrates statistically significant differences in the β values of the left and right visual cortex within the occipital lobe during the task state HbO_2 between pre- and post-cylindrical correction conditions ($P = 0.002$, $P = 0.003$).

Changes in brain function with different axial astigmatism defocus in the task state

In the third section of the study, 24 participants (48 eyes) underwent induced astigmatism at various axes using +2.00 diopter cylindrical lenses, following the correction of their original refractive errors. Based on the axial orientation, the participants were categorized into three groups: the WTR group, comprising 8 individuals with 16 eyes (+2.00DC $\times$ 90°); the oblique astigmatism group, consisting of 8 individuals with 16 eyes (+2.00DC $\times$ 45°); and the ATR group, also with 8 individuals and 16 eyes (+2.00DC $\times$ 180°). The baseline features of these groups are detailed in Table 3. All participants underwent testing in both eyes, beginning with the right eye followed by the left.

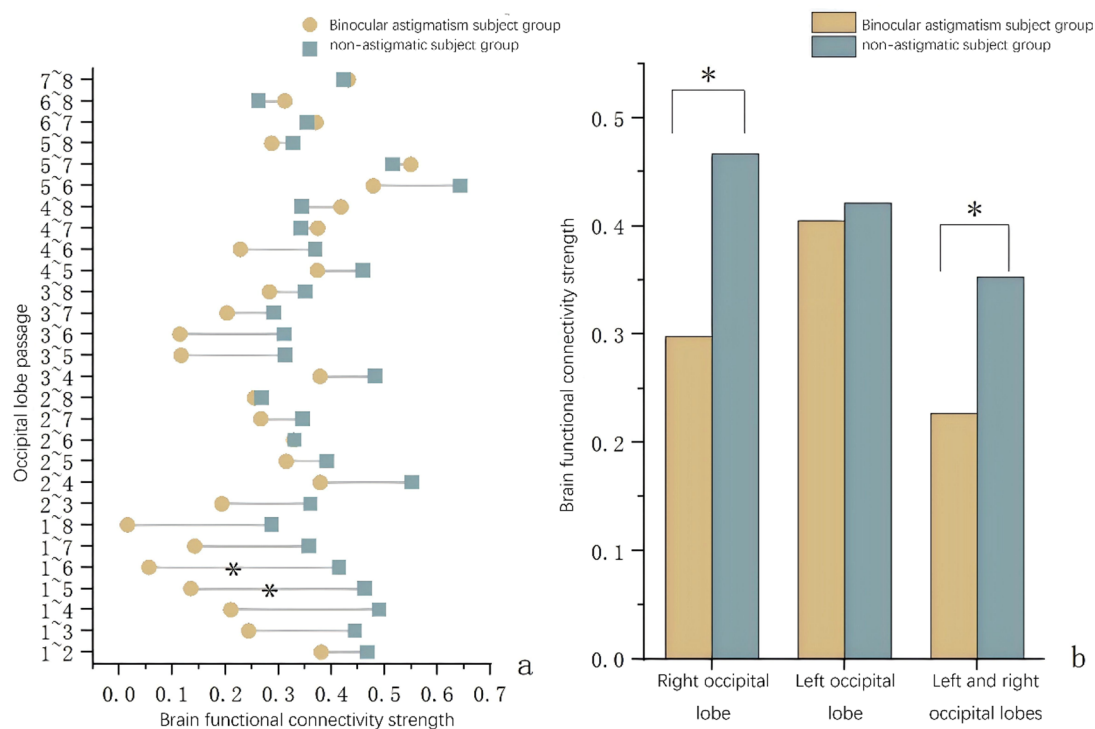


Fig. 5. The study conducted a comparative analysis of brain functional connectivity strength, focusing on individual channels (a) and specific brain regions (b) within the visual cortex of the occipital lobe, in relation to HbO₂ levels during the resting state. Group A comprised individuals with binocular astigmatism, while Group B consisted of individuals without astigmatism. * $P < 0.05$.

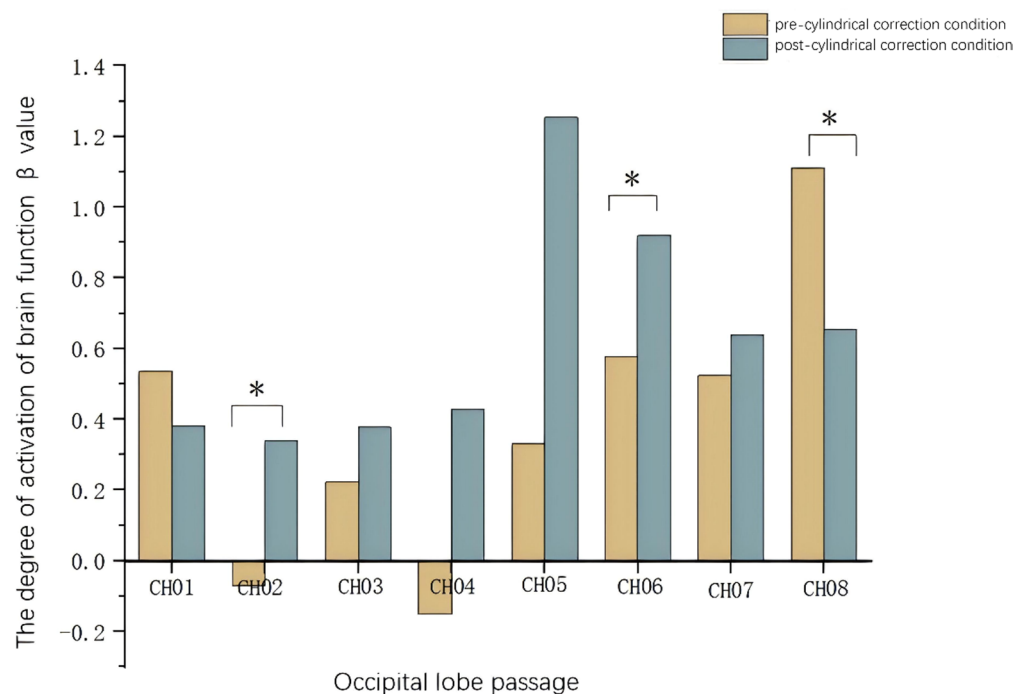


Fig. 6. The β values for each channel of the occipital lobe under the task state HbO₂ were compared, with pre-cylindrical correction condition representing participants prior to astigmatism correction and post-cylindrical correction condition representing participants following astigmatism correction, * $P < 0.05$.

	Pre-cylindrical correction condition	Post-cylindrical correction condition	t-value	P-value
Right	0.13 ± 0.70	0.38 ± 0.78	-2.03	0.047
Left	0.64 ± 0.96	0.77 ± 0.98	-0.79	0.435
t-value	-3.27	-2.96		
P-value	0.002	0.004		

Table 2. The β values of the visual cortex on the left and right sides of the occipital lobe of the brain based on HbO_2 .

	WTR group	Oblique astigmatism group	ATR group	P-value
Gender				0.68
Male	5	3	3	
Female	3	5	5	
Number of eyes/cases	16	16	16	
Age/years	20.50 ± 0.87	21.75 ± 0.65	20.75 ± 0.75	0.63
intraocular pressure /mmHg	16.38 ± 0.42	15.75 ± 0.62	16.25 ± 0.82	0.85
Spherical power /DS	-3.63 ± 0.92	-3.88 ± 1.38	-3.28 ± 1.11	0.35
Cylindrical degree /DC	-1.16 ± 0.41	-1.36 ± 0.47	-1.36 ± 0.40	0.30

Table 3. Comparison of baseline data.

As illustrated in Fig. 7, a statistically significant variation was detected in the strength of brain functional connectivity associated with the HbO_2 occipital visual cortex across different astigmatism axes ($P < 0.001$). The Bonferroni method was employed for post-hoc comparisons between the groups. The results indicated that the oblique astigmatism group exhibited higher brain functional connectivity than both the WTR group ($P = 0.001$) and the ATR group ($P < 0.001$). Furthermore, the intensity of cerebral functional connectivity was markedly higher in the oblique astigmatism group compared to the ATR group ($P = 0.019$). Figure 8 and 9 illustrates the strength of three axial brain functional connections across each channel and brain region. As depicted in Fig. 8 and 9a, statistically significant differences among the three groups were observed in the following connections: CH01-CH05 ($P = 0.012$), CH02-CH05 ($P = 0.042$), CH05-CH06 ($P = 0.049$), and CH07-CH08 ($P = 0.003$). Additionally, a statistically significant disparity in the CH01-CH05 connection was detected between the WTR group and the oblique astigmatism group ($P = 0.047$). Significant statistical variations were discerned between the ATR group and the oblique astigmatism group in the comparisons CH01-CH05 ($P = 0.031$), CH02-CH05 ($P = 0.048$), and CH07-CH08 ($P = 0.002$). Conversely, no notable variations were detected between the ATR group and the WTR group. As illustrated in Fig. 9b, a notable statistical discrepancy was present in the strength of functional connectivity among the three groups within the right occipital lobe ($P = 0.012$) and the left occipital lobe ($P = 0.046$) across homologous brain networks. The Bonferroni method was employed to conduct intergroup comparisons of brain function, revealing that the functional connectivity strength within the right occipital homologous brain network was markedly higher in the oblique astigmatism group relative to the ATR group, with the difference reaching statistical significance ($P = 0.049$). In the homology brain network of the left occipital lobe, the intensity of functional connectivity was significantly higher in the oblique astigmatism group compared to that in the ATR group ($P < 0.001$) and the WTR group ($P = 0.001$), with the differences between these groups being statistically significant ($P < 0.001$). A statistically significant difference in brain functional connectivity strength was present among the three groups ($P < 0.001$). The Bonferroni method was employed for pairwise comparisons, revealing that the oblique astigmatism group exhibited higher functional connectivity than the ATR group, with this difference also reaching statistical significance ($P = 0.011$).

Figure 10 illustrates that there were statistically significant differences in occipital visual cortex activation, as measured by HbO_2 , across different astigmatism axes ($P = 0.045$). The Bonferroni method was employed to compare brain activation between the two groups, revealing that the ATR group exhibited a higher degree of brain activation compared to the oblique astigmatism group. This difference was statistically significant ($P = 0.042$). The variation in β among the three groups was statistically significant in channel CH02 ($P = 0.022$), with the β value of the ATR astigmatism group exceeding that of the oblique astigmatism group ($P = 0.019$).

Discussion

This study represents the inaugural application of fNIRS technology to assess the strength of brain functional connectivity associated with astigmatism. It compares alterations in visual cortex brain function before and after astigmatism correction, as well as between the left and right hemispheres in subjects with monocular astigmatism. Additionally, it examines the changes in visual cortex brain function induced by lenses with varying astigmatism axes oriented downward. Gao et al.³⁹ investigated the impact of astigmatism on visual quality in children using the OPD-SCAN III aberrometer. Their results showed that as the degree of astigmatism increased, the root mean square values of total higher-order aberrations, total coma, total spherical aberration,

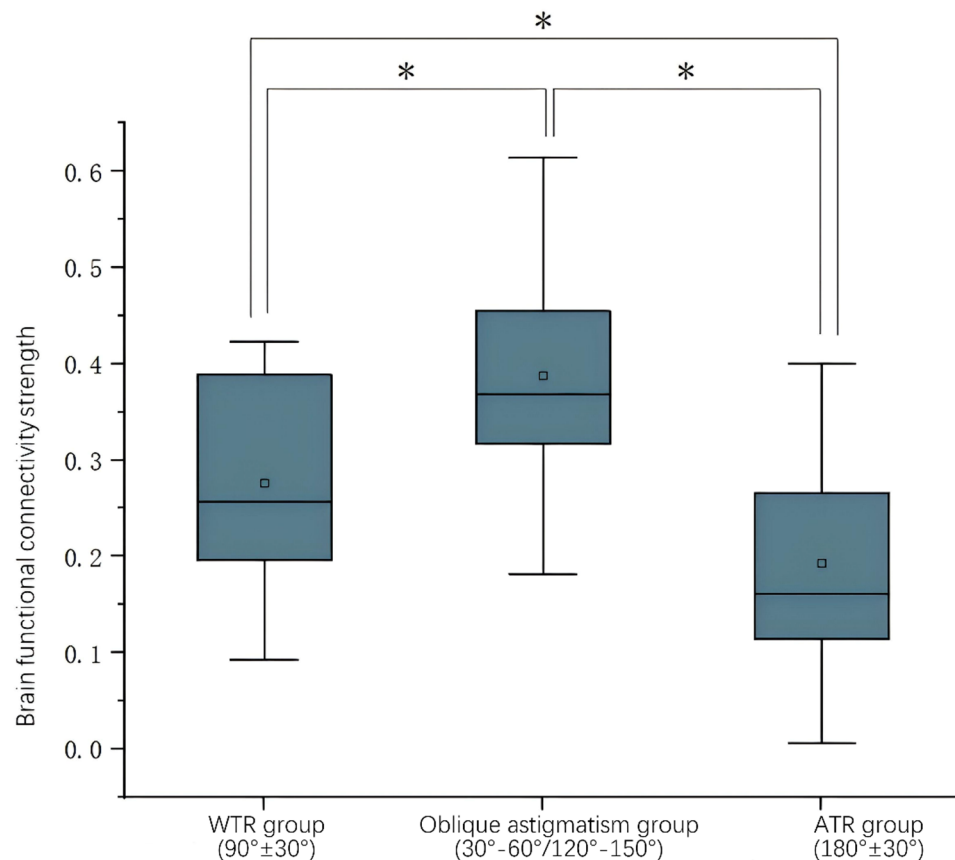


Fig. 7. Comparison of brain functional connectivity strength in each axial group of occipital visual cortex based on HbO_2 in the task state. The small square indicates the mean; the line indicates the median, $*P < 0.05$.

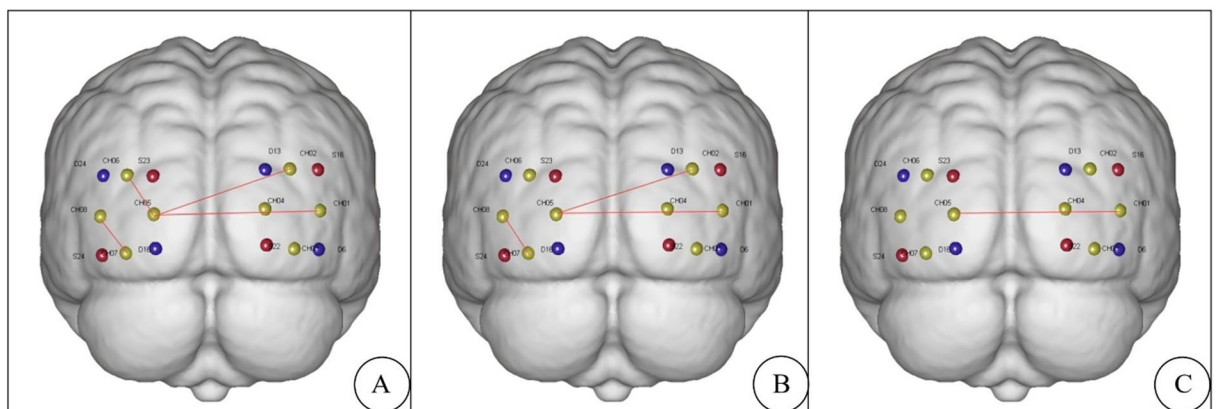


Fig. 8. Channels with significant group differences. (A) CH01-CH05, CH02-CH05, CH05-CH06, and CH07-CH08 showed differences among the three groups ($P < 0.05$). (B) CH01-CH05 showed difference between the WTR and oblique astigmatism groups ($P < 0.05$). (C) CH01-CH05, CH02-CH05, and CH07-CH08 showed difference between the ATR and oblique astigmatism groups ($P < 0.05$). The red line represents $P < 0.05$.

and total trefoil all increased, while the modulation transfer function (MTF) and point spread function (PSF) decreased. The findings of the present study demonstrate that MTF cut off, strehl ratio (SR), VA100%, VA20%, and VA9% all decrease with increasing astigmatic power. Under the same degree of astigmatism, there were no statistically significant differences in these parameters across different axial orientations.

The findings indicated that during the initial phase of the resting-state HbO_2 occipital visual cortex functional connectivity assessment, individuals with astigmatism exhibited reduced functional connectivity strength in the occipital visual cortex compared to those without astigmatism. This suggests that prolonged abnormal visual

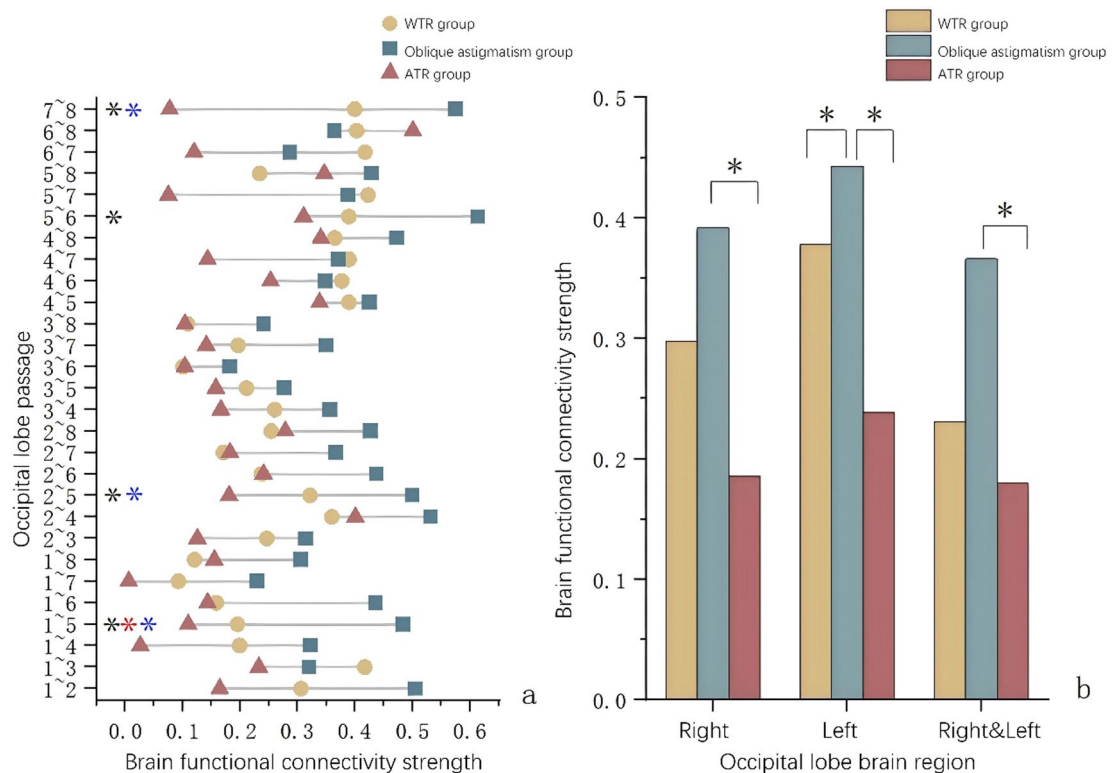


Fig. 9. Comparison of brain functional connectivity strength of each channel (a) and each brain region (b) of the occipital visual cortex of the brain based on three sets of axial task states. In Fig. 9a, * (red) represents the channel with statistically significant differences among the three groups ($P < 0.05$), * (red) represents the channel exhibiting a statistically significant difference between the two groups with a WTR group and an oblique astigmatism group, and * (blue) represents a channel exhibiting a statistically significant difference between the ATR group and the oblique astigmatism group. In Fig. 9b, * $P < 0.05$.

stimulation associated with astigmatism may result in diminished functional connectivity between the occipital visual cortices.

Consistent with the animal literature highlighting the vulnerability of the visual cortex to abnormal early visual input^{6,12}, the present study provides neuroimaging evidence in humans that astigmatism is associated with cortical functional changes. While previous research has primarily focused on the optical components of astigmatism³⁹, our fNIRS findings suggest that the neural impact extends to the visual cortex, particularly in adults with a history of prolonged distorted visual input during development. The persistent shape perception biases observed behaviorally in adults with astigmatism, even after optical correction²³, imply a neural rather than purely optical origin. Our results offer a potential neural correlate for this phenomenon: the aberrant cortical activation patterns detected by fNIRS may reflect the long-term adaptation of the visual system to chronic asymmetric or degraded input. It is plausible that these cortical changes represent the neural substrate underlying the incomplete recovery of perceptual function following correction. Although the current study did not directly measure behavioral performance, the observed cortical changes in adults with astigmatism may provide insight into the neural basis of these persistent perceptual deficits. Future investigations integrating fNIRS with concurrent behavioral assessments (e.g., contrast sensitivity, orientation discrimination, or shape perception tasks) are warranted to establish direct correlations between neural activation patterns and visual performance, thereby elucidating the precise relationship between cortical function and perceptual behavior in astigmatism.

Furthermore, our research team investigated the impact of astigmatism correction on alterations in brain function. We discovered that correcting astigmatism results in a clearer retinal image, enhances the retinal signal transmission to the visual cortex, and ultimately leads to increased neural activity and alterations in blood oxygen levels within the visual cortex. In the task-based HbO₂ occipital visual cortex brain functional connectivity assessment, No significant statistical difference was detected in the strength of brain functional connectivity pre- and post-astigmatism correction. However, a general trend of increased connectivity strength post-correction was noted. In the brain function activation assessment utilizing task-state HbO₂ measurements in the occipital visual cortex, the β value post-astigmatism correction was significantly elevated compared to pre-correction levels. Notably, this difference was statistically significant in channels CH02, CH06, and CH08, suggesting that astigmatism correction enhances the functional activity of the occipital visual cortex to a certain extent.

Conversely, to investigate potential differences in the degree of activation within the bilateral occipital visual cortex among individuals with astigmatism, the second segment of this study employed monocular

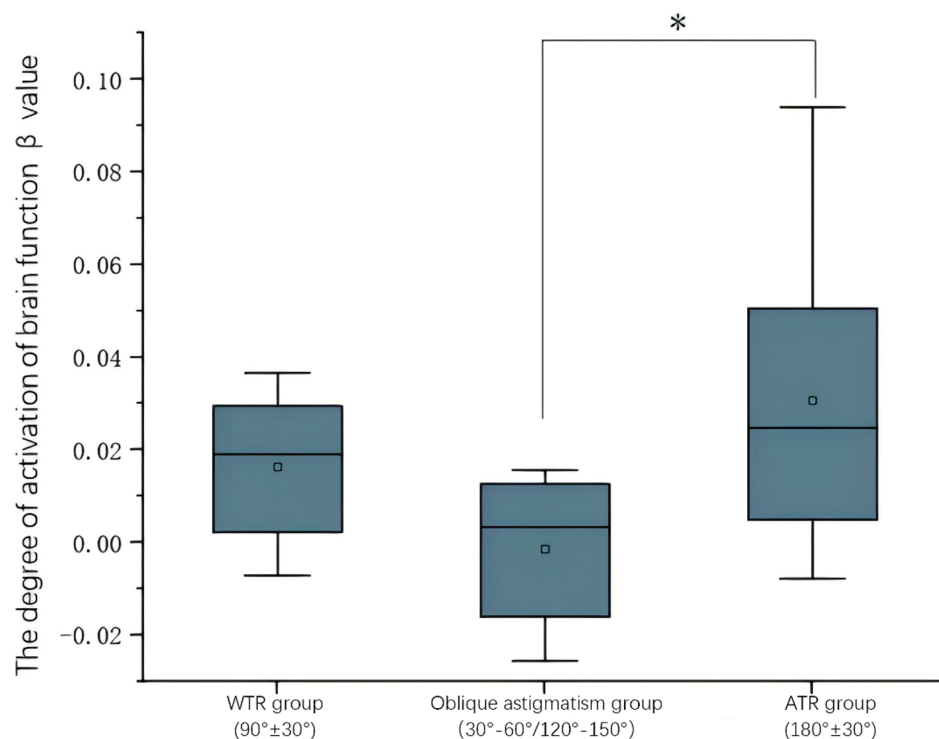


Fig. 10. Comparison of brain activation degree based on HbO₂ occipital visual cortex in the three groups of axial task states. The small square indicates the mean; the line indicates the median, * $P < 0.05$

visual stimulation in participants with monocular astigmatism. The findings indicated that the β value of the ipsilateral visual cortex corresponding to the stimulated eye was lower than that of the contralateral visual cortex in subjects with astigmatism, both prior to and following astigmatism correction. The transmission of visual information from the retina to the visual cortex exhibits asymmetry. Specifically, 47% of the information from the temporal retina is conveyed to the ipsilateral visual cortex, while 53% of the information from the nasal retina is transmitted to the contralateral visual cortex⁴⁰. Variations in the activation levels of the occipital visual cortex have been observed in individuals with monocular astigmatism. These variations may be attributed to prolonged differential visual stimulation of the bilateral retina, leading to asymmetrical alterations in brain function between the bilateral visual cortices.

Astigmatism, being a vector quantity, influences visual quality not solely based on its magnitude but also in relation to its axis orientation. Mimouni et al.⁴¹ conducted an evaluation of the impact of axial astigmatism on VA in patients exhibiting residual astigmatism following refractive surgery. Their findings indicated that the influence of ondate astigmatism was less pronounced compared to that of retrograde astigmatism and oblique axial astigmatism. The influence of the astigmatism axis seems to be associated with the "tilt effect," wherein visual stimuli exhibit greater resolving power along the horizontal and vertical axes compared to the oblique axis⁴². This phenomenon has been corroborated by studies employing adaptive optics¹² and fMRI⁴³, which suggest a connection to neural mechanisms within the visual cortex. Furthermore, it is hypothesized that this effect may be linked to the directional sensitivity of retinal ganglion cells⁸. Astigmatism exhibits a neuroadaptive phenomenon whereby the human visual system, when subjected to persistent astigmatism and defocus, can adapt to the resultant blurring. This adaptation leads to enhancements in both VA and visual quality². Clinically, this phenomenon has been observed to exhibit selectivity concerning the axis of astigmatism, with oblique axis astigmatism proving more challenging to adapt to compared to compliant and retrograde astigmatism. Therefore, the impact of the axial direction of astigmatism was further examined across different groups in our study. In examining the strength of brain functional connectivity associated with the occipital visual cortex of the occipital lobe during task-state HbO₂ brain activity, statistically significant differences were observed among patients with compliant, oblique axis, and inverse astigmatism. Notably, the strength of brain functional connectivity in the oblique astigmatism group was greater than that observed in both the oblique astigmatism group and the antigrade astigmatism group. The findings suggest that the strength of functional brain connectivity varies with different astigmatic axial stimulations. This variation may be attributed to the blurring effects induced by lens-induced oblique axis astigmatism, which poses greater adaptive challenges compared to the other two axes. Oblique axis astigmatism demands heightened attention and visual processing, necessitating the recruitment of a broader array of brain regions to effectively integrate visual feedback. Abnormal visual experiences resulting from astigmatism can disrupt brain development, leaving "traces" in the adult visual cortex. This phenomenon may be associated with the diminished collinear facilitation effect observed in oblique axis astigmatism, which arises from interactions among optic neurons and can be regarded as an intermediate stage in the processing of

visual information²². Elliott et al.⁴⁴ proposed that the increased difficulty in adapting to oblique axis astigmatism may be associated with vertical visual perception, which is believed to be mediated by the occipital cortical network. In assessments of vertical visual perception using a computer-based subjective visual vertical task, the subjective visual vertical error in individuals with oblique axis astigmatism was found to be greater than that observed in individuals with regular or anti-regular astigmatism. However, emerging evidence indicates that afferent and efferent neural pathways between the brain and the eye do not contribute to the induction of astigmatism, whereas the degeneration of retinal cells has been shown to disrupt the development of this refractive error¹.

In this investigation, we completed a comparative analysis of the β values representing brain activation in the occipital visual cortex, as measured by HbO₂, across different task states. The findings of our research exhibited a statistically significant variation in the degree of brain activation among patients with oblique axis astigmatism, oblique axis astigmatism, and retrograde astigmatism. Specifically, the group with oblique axis astigmatism exhibited a lower degree of brain activation compared to the ATR group. These results suggest that the activation level of the visual cortex varies according to the axis of astigmatism. Neurodeficits in the V1 region of the visual cortex have been documented in both animal experiments and human psychophysical studies concerning astigmatism-induced meridian amblyopia²¹. The primary visual cortex cells exhibit orientation selectivity, with neurons in the V1 region demonstrating a predisposition towards horizontal and vertical orientations over inclined ones⁴⁵. For example, Gu et al.²⁰ employed a combination of psychophysical and electroencephalogram methodologies to investigate contrast sensitivity and visual evoked potentials along the two principal meridians in children without astigmatism and those with meridian amblyopia. Their study confirmed the horizontal and vertical asymmetry associated with human astigmatism in spatial vision and identified direction-specific visual developmental deficits in the primary visual cortex. Additionally, Shiba et al.⁴⁶ conducted a study involving refractive correction on two subjects, followed by the administration of +6.00 diopters of compliant and retrograde astigmatism defocus. fMRI was utilized to assess the activation levels of the visual cortex during visual stimulation. The findings indicated that astigmatism potentially influences the human visual cortex, with variations in activation levels observed across different axes.

However, this study has some limitations. Primarily, the investigation was confined to the occipital visual cortex, leaving the involvement of functional changes in other brain regions undetermined. The exclusive use of fNIRS, while providing valuable hemodynamic data, limits our ability to fully characterize neural mechanisms. The absence of complementary techniques like EEG (for temporal resolution) or fMRI (for spatial precision) means we cannot determine whether observed hemodynamic changes reflect altered neural timing or specific subcortical contributions. Although we included measures of VA (Log MAR) and contrast sensitivity (Pelli-Robson), our study lacked behavioral assessments related to visual function, such as tests of visual perception, dynamic acuity, or real-world functional vision. Additionally, the study's broader development was constrained by the operational complexity of the testing equipment and the high level of participant cooperation required. While we reported functional connectivity differences between astigmatism subgroups, the absence of repeated fNIRS measurements in the same subjects prevented us from performing additional validation analyses such as test–retest reliability assessment. And the relatively small sample size in certain subgroups, which could reduce the robustness of signal quality comparisons. Furthermore, this study included individuals with monocular astigmatism and those with binocular astigmatism. However, a direct comparison between these groups was not feasible due to the experimental design. The monocular astigmatism group was assessed using a within-subject design (comparing cortical activation before and after correction under monocular viewing), whereas the binocular astigmatism group was assessed using a between-subject design (comparing with healthy controls under binocular viewing). Directly comparing these two groups would confound the effects of astigmatism type with differences in viewing conditions. Future studies employing a unified protocol are needed to systematically compare these two refractive conditions. This remains a limitation of the present work. In addition, the sample selection was relatively limited, as participants with a broader range of spherical refractive errors combined with astigmatism were not included for further analysis.

Future research should pursue three key directions to advance understanding of astigmatism-related cerebral functional changes. First, expanded sample sizes with balanced representation across astigmatism subtypes are needed to clarify their differential neural effects. Second, multimodal integration of fNIRS with fMRI technology will enable simultaneous, validated assessment of visual pathway activation patterns with complementary spatial and temporal resolution. Finally, establishing standardized protocols specifically for astigmatic populations—including rigorous test–retest reliability assessments, refractive-error-adapted signal quality metrics, and individualized optode placement strategies—will be critical for enhancing methodological rigor. These coordinated advances will substantially improve both the reproducibility and clinical applicability of neuroimaging findings in visual impairment research, ultimately facilitating more precise characterization of how different astigmatism subtypes influence brain function. This approach will facilitate the acquisition of more comprehensive and reliable data, thereby providing a deeper understanding of the changes and patterns in brain function among patients with clinical astigmatism.

Conclusion

The present findings suggest that uncorrected astigmatism may lead to reduced activation in the occipital visual cortex and weakened brain functional connectivity. Notably, optical correction of astigmatism appears to enhance the functional activity of the occipital cortex, at least partially, highlighting the potential neuroplastic effects of vision restoration. Furthermore, the degree of cortical activation and the strength of functional connectivity vary depending on the axis of astigmatism, implying that the neural impact of astigmatism may be orientation-dependent. However, these connectivity differences were derived from fNIRS measurements, and further research is needed to validate the sensitivity and reliability of this neuroimaging modality in populations

with astigmatism. Future studies should explore the long-term neural adaptations to astigmatism correction and investigate the underlying mechanisms linking optical aberrations to functional brain reorganization.

Data availability

All data generated or analyzed during this study are included in this published article.

Received: 18 November 2025; Accepted: 1 April 2026

Published online: 08 April 2026

References

1. Popa, A. V., Kee, C. S. & Stell, W. K. Retinal control of lens-induced astigmatism in chicks. *Exp. Eye Res.* **194**, 108000 (2020).
2. Leung, T. W., Li, R. W. & Kee, C. S. Brief adaptation to astigmatism reduces meridional anisotropy in contrast sensitivity. *Invest. Ophthalmol. Vis. Sci.* **64**(12), 4 (2023).
3. Wang, M. et al. Prevalence of and risk factors for refractive error: A cross-sectional study in Han and Mongolian adults aged 40–80 years in Inner Mongolia, China. *Eye* **33**(11), 1722–1732 (2019).
4. Sanfilippo, P. G. et al. Distribution of astigmatism as a function of age in an Australian population. *Acta Ophthalmol.* **93**(5), e377–e385 (2015).
5. Son, S., Shim, W. M., Kang, H. & Lee, J. Automatic compensation enhances the orientation perception in chronic astigmatism. *Sci. Rep.* **12**(1), 3710 (2022).
6. Mitchell, D. E., Freeman, R. D., Millodot, M. & Haegerstrom, G. Meridional amblyopia: Evidence for modification of the human visual system by early visual experience. *Vision Res.* **13**(3), 535–558 (1973).
7. Aykut, V., Kirgiz, A., Gül Ay, B. & Celik, U. Comparison of pre-incision and single-stepped clear corneal incision in phacoemulsification surgery. *Eur. Rev. Med. Pharmacol. Sci.* **18**(12), 1698–1703 (2014).
8. Vyas, S. A., Lakshmanan, Y., Chan, H. H., Leung, T. W. & Kee, C. S. Experimentally induced myopia and myopic astigmatism alter retinal electrophysiology in chickens. *Sci. Rep.* **12**(1), 21180 (2022).
9. Kulp, M. T. et al. Vision screening, vision disorders, and impacts of hyperopia in young children: outcomes of the vision in preschoolers (VIP) and vision in pre-schoolers—hyperopia in preschoolers (VIP-HIP) studies. *Asia Pac. J. Ophthalmol.* **11**(1), 52–58 (2022).
10. Vinas, M., Sawides, L., de Gracia, P. & Marcos, S. Perceptual adaptation to the correction of natural astigmatism. *PLoS ONE* **7**(9), e46361 (2012).
11. Yap, T. P., Luu, C. D., Suttle, C. M., Chia, A. & Boon, M. Y. Electrophysiological and psychophysical studies of meridional anisotropies in children with and without astigmatism. *Invest. Ophthalmol. Vis. Sci.* **60**(6), 1906–1913 (2019).
12. Zheleznyak, L., Barbot, A., Ghosh, A. & Yoon, G. Optical and neural anisotropy in peripheral vision. *J. Vis.* **16**(5), 1 (2016).
13. Agrawal, A. et al. Alteration in pattern reversal visual evoked potential in pediatric population with amblyopia and spherical refractive errors. *Indian J. Ophthalmol.* **70**(5), 1727–1731 (2022).
14. Tang, W. Y. et al. Application of the virtual reality-pattern visual evoked potential in forensic visual acuity evaluation. *Fa Yi Xue Za Zhi* **36**(6), 762–766 (2020).
15. Suzuki, M. et al. Effects of refractive errors on visual evoked magnetic fields. *BMC Ophthalmol.* **15**, 162 (2015).
16. Cheng, Y. et al. Abnormal functional connectivity between cerebral hemispheres in patients with high myopia: A resting fMRI study based on voxel-mirrored homotopic connectivity. *Front. Hum. Neurosci.* **16**, 910846 (2022).
17. Ji, Y. et al. Exploration of static functional connectivity and dynamic functional connectivity alterations in the primary visual cortex among patients with high myopia via seed-based functional connectivity analysis. *Front. Neurosci.* **17**, 1126262 (2023).
18. Freeman, R. D., Mitchell, D. E. & Millodot, M. A neural effect of partial visual deprivation in humans. *Science* **175**(4028), 1384–1386 (1972).
19. Harvey, E. M., Dobson, V., Miller, J. M. & Clifford-Donaldson, C. E. Changes in visual function following optical treatment of astigmatism-related amblyopia. *Vision Res.* **48**(6), 773–787 (2008).
20. Gu, L. et al. Meridian-specific and post-optical deficits of spatial vision in human astigmatism: Evidences from psycho-physical and EEG scalings. *Front. Psychol.* **12**, 595536 (2021).
21. Yap, T. P., Luu, C. D., Suttle, C., Chia, A. & Boon, M. Y. Effect of stimulus orientation on visual function in children with refractive amblyopia. *Invest. Ophthalmol. Vis. Sci.* **61**(5), 5 (2020).
22. Serero, G., Lev, M. & Polat, U. Meridional binocular rivalry reveals a trace of uncorrected oblique input during development in the adult brain. *Sci. Rep.* **13**(1), 9920 (2023).
23. Serero, G., Lev, M., Sagi, D. & Polat, U. Traces of early developmental bias in the adult brain. *Sci. Rep.* **13**(1), 12014 (2023).
24. Vincent, M., Marin, G. & Legras, R. Effect of simulated and real spherical and astigmatism defocus on visual acuity and image quality score. *Optom. Vis. Sci.* **97**(1), 36–44 (2020).
25. Liang, D., Leung, T. W. & Kee, C. S. Measuring retinal thickness and visual acuity in eyes with different types of astigmatism in a cohort of Hong Kong Chinese adults. *Invest. Ophthalmol. Vis. Sci.* **64**(1), 2 (2023).
26. Tan, Q. Q. et al. Optical quality in low astigmatic eyes with or without cylindrical correction. *Graefes Arch. Clin. Exp. Ophthalmol.* **258**(2), 451–458 (2020).
27. Pinti, P. et al. The present and future use of functional near-infrared spectroscopy (fNIRS) for cognitive neuroscience. *Ann. N. Y. Acad. Sci.* **1464**(1), 5–29 (2020).
28. Fishburn, F. A., Norr, M. E., Medvedev, A. V. & Vaidya, C. J. Sensitivity of fNIRS to cognitive state and load. *Front. Hum. Neurosci.* **8**, 76 (2014).
29. Dong, Y. et al. Frontal and parietal cortices activation during walking is repeatable in older adults based on fNIRS. *Heliyon* **10**(9), e30197 (2024).
30. Yin, X. et al. Altered brain structure and spontaneous functional activity in children with concomitant strabismus. *Front. Hum. Neurosci.* **15**, 777762 (2021).
31. Wang, Y., Wu, Y., Luo, L. & Li, F. Structural and functional alterations in the brains of patients with anisometropic and strabismic amblyopia: A systematic review of magnetic resonance imaging studies. *Neural Regen. Res.* **18**(11), 2348–2356 (2023).
32. Aitchison, R. T. et al. Measuring visual cortical oxygenation in diabetes using functional near-infrared spectroscopy. *Acta Diabetol.* **55**(11), 1181–1189 (2018).
33. Zhang, Y. et al. Changes in visual cortical function in moderately myopic patients: A functional near-infrared spectroscopy study. *Ophthalmic Physiol. Opt.* **42**(1), 36–47 (2022).
34. Baller, E. B. et al. Developmental coupling of cerebral blood flow and fMRI fluctuations in youth. *Cell Rep.* **38**(13), 110576 (2022).
35. Wagner, J. C. et al. Reproducibility of fNIRS within subject for visual and motor tasks. *Neuroimage*. **321**(1), 121492. (2025).
36. Gao, C., Xiu, J., Huang, C., Ma, K. & Li, T. Reliability evaluation for continuous-wave functional near-infrared spectroscopy systems: Comprehensive testing from bench characterization to human test. *Sensors*. **24**(7), (2024).
37. Dubois, J. et al. Reliability of brain metrics derived from a time-domain functional near-infrared spectroscopy system. *Sci. Rep.* **14**(1), 17500 (2024).

38. Odom, J. V. et al. ISCEV standard for clinical visual evoked potentials: (2016 update). *Doc. Ophthalmol.* **133**(1), 1–9 (2016).
39. Gao, J. et al. The effect of the degree of astigmatism on optical quality in children. *J. Ophthalmol.* **2017**, 5786265 (2017).
40. Prasad, S. & Galetta, S. L. Anatomy and physiology of the afferent visual system. *Handb. Clin. Neurol.* **102**, 3–19 (2011).
41. Mimouni, M. et al. The effect of astigmatism axis on visual acuity. *Eur. J. Ophthalmol.* **27**(3), 308–311 (2017).
42. Li, B., Peterson, M. R. & Freeman, R. D. Oblique effect: A neural basis in the visual cortex. *J. Neurophysiol.* **90**(1), 204–217 (2003).
43. Furmanski, C. S. & Engel, S. A. An oblique effect in human primary visual cortex. *Nat. Neurosci.* **3**(6), 535–536 (2000).
44. Elliott, D. B., Black, A. A. & Wood, J. M. Subjective verticality is disrupted by astigmatic visual distortion in older people. *Invest. Ophthalmol. Vis. Sci.* **61**(4), 12 (2020).
45. Fang, C., Cai, X. & Lu, H. D. Orientation anisotropies in macaque visual areas. *Proc. Natl. Acad. Sci. U. S. A.* **119**(15), e2113407119 (2022).
46. Shiba, T., Tsuneoka, H. & Jiang, Z. Influence of regular astigmatism on the human visual cortex. A functional magnetic resonance imaging study. *J. Ophthalmic Sci.* **1**(1), 28–38 (2015).

Author contributions

YLT and CW conceived the study, organized the data collection, and designed the data analysis process. CW, MTB, QQT, JT, JL, XYC, MXL and ZWC performed the data collection. YLT and CW performed the data analysis. TYL and WC wrote the manuscript. YS and XL supervised the project. All authors contributed to the scientific discussion and manuscript revisions.

Funding

Natural Science Foundation Project of Sichuan Provincial Department of Science and Technology 2023NS-FSC0595, Key Project of Scientific Research Development Plan of the Affiliated Hospital of North Sichuan Medical College 2023ZD010.

Declarations

Competing interests

The authors declare that they have no competing interests.

Additional information

Correspondence and requests for materials should be addressed to Y.S. or X.L.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

© The Author(s) 2026